Problem Description

The **P2P Shopping Assistant Application** is a hybrid mobile/web application that helps users manage shopping lists while leveraging crowd-sourced indoor positioning data (GPS + Wi-Fi RTT) to dynamically map product locations inside supermarkets.

When users mark an item as completed, the system records their location and aggregates this information across multiple users to create a dynamic product-location map. Future users benefit from optimized routing and store suggestions.

The system also supports shared shopping lists, real-time collaboration, and route optimization (Multi-Agent TSP).

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1. Actors

Primary Actors

- **Guest User** – Can register or log in
- **Registered User** – Manages account and shopping lists
- **Active Shopper** *(extends Registered User)* – In-store user with GPS/Wi-Fi RTT active
- **List Collaborator** – Co-edits shared list in real-time
- **Administrator** – Manages users and system data

Secondary (System) Actors

- **GPS / Location Service** – Provides real-time coordinates
- **Map Aggregation Server** – Collects and clusters item locations
- **Notification Service** – Sends alerts and updates

2. Use Cases

Authentication

- **UC1 – Manage Account** (Guest, Registered User)

List Management

- **UC2 – Manage List** (create / edit / delete items)
- **UC3 – Set Item Recurrent**
- **UC4 – View Shopping History**
- **UC5 – Mark Item Completed**
    - «includes» UC6

P2P Mapping

- **UC6 – Record Item Location** (Active Shopper, GPS)
    - «includes» UC7
- **UC7 – Contribute Location Data** (Map Server)
- **UC11 – View Dynamic Product Map**
- **UC12 – Suggest Best Supermarket**
    - «includes» UC11
- **UC13 – Get Optimal Route (TSP)**
    - «includes» UC11
- **UC14 – Navigate to Item**
    - «extends» UC13
- **UC15 – Alert Near Item**
    - «extends» UC14

Collaboration

- **UC8 – Share List**
  - «extends» UC9
- **UC9 – Invite Collaborator**
- **UC10 – Edit Shared List (Real-Time)**
  - «extends» UC8

Administration

- **UC16 – Manage Users**
- **UC17 – Moderate Location Data**
- **UC18 – Map Analytics**

Use-Case Relations

«include»

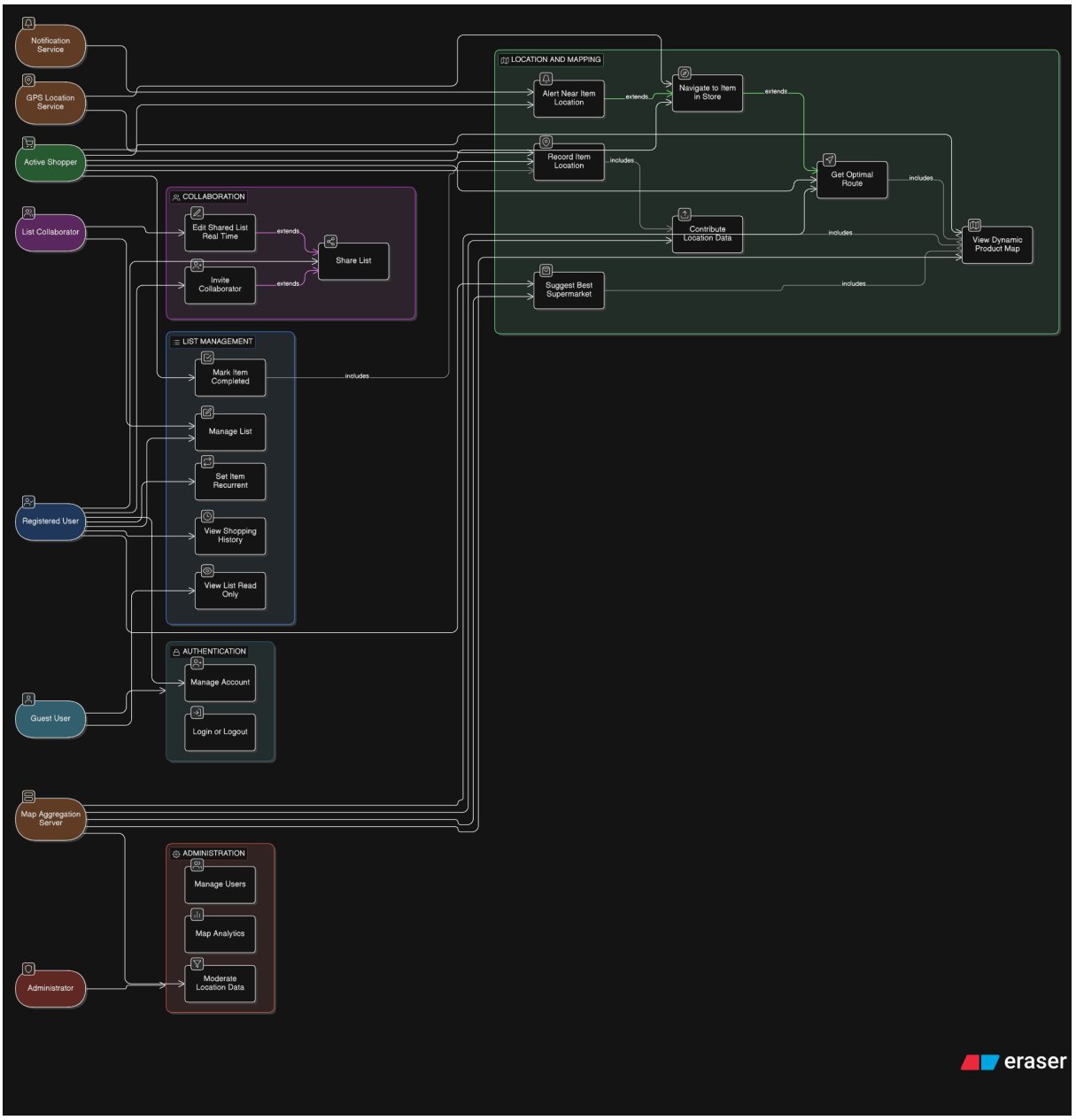
- UC5 → UC6
- UC6 → UC7
- UC12 → UC11
- UC13 → UC11

«extend»

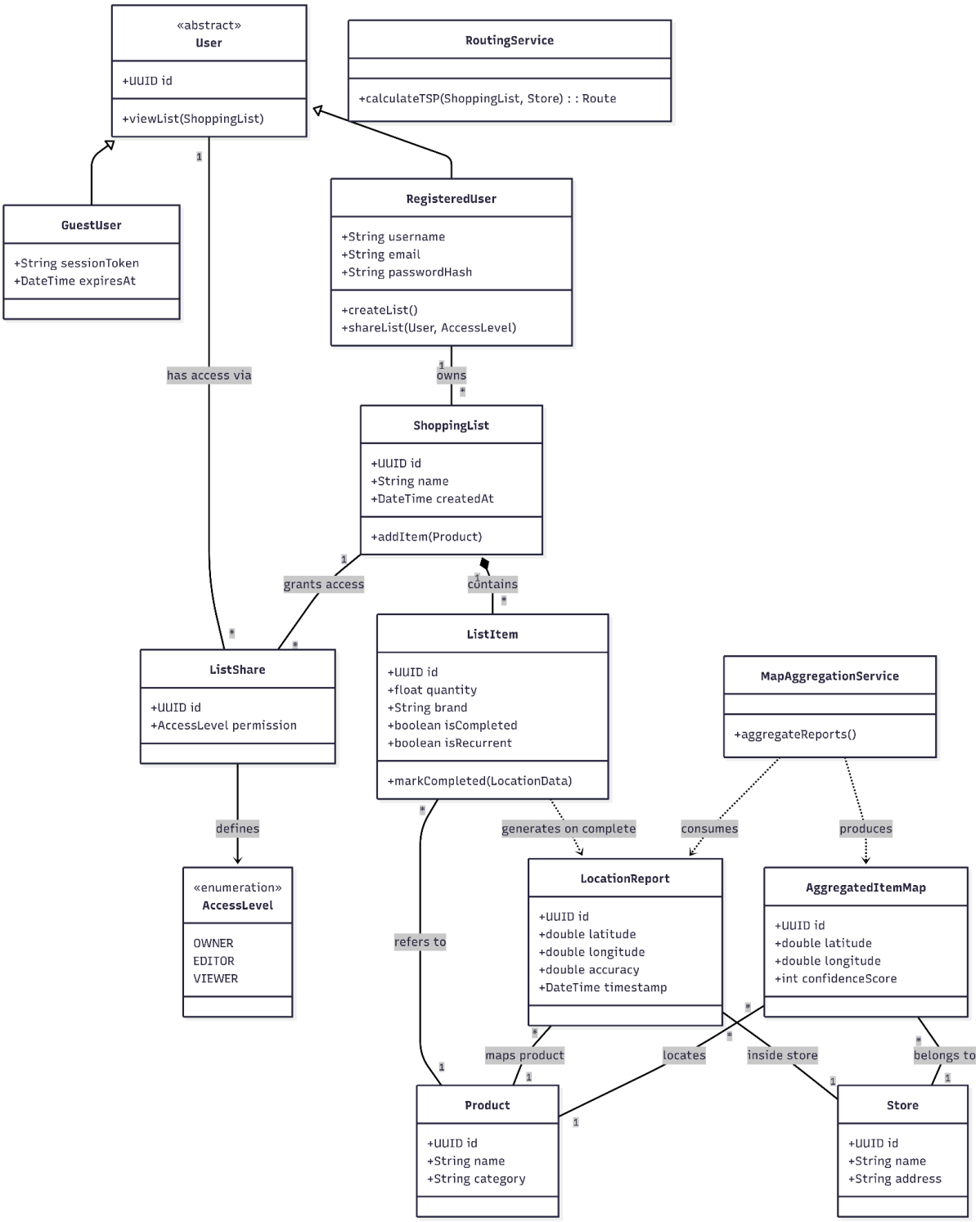
- Active Shopper → Registered User (generalization)
- UC14 → UC13

- UC15 → UC14
- UC8 → UC9
- UC10 → UC8

Use-Case Diagram



Class Diagram



AI usage:

-ChatGPT (for problem description and other small aspects)

Prompt used: “We were given a group project for our entire class of about 30 students. Here are the details: (inserted entire project description), Here are the tasks we were given, please organize everything solved neatly:(inserted the entire text from the lab 3 document)”

-DiagramGPT (for the use case diagram)

Prompt used:”Create a UML Use Case diagram for a P2P Shopping mobile application.

System boundary: "P2P Shopping Application"

Actors (left side): Guest User, Registered User, Active Shopper, List Collaborator

Actors (right side): Administrator, GPS/Location Service, Map Aggregation Server, Notification Service

Use Cases:

AUTH: Manage Account

LIST MANAGEMENT: Manage List, Set Item Recurrent, View Shopping History, Mark Item Completed

COLLABORATION: Share List, Invite Collaborator, Edit Shared List (Real-Time)

P2P LOCATION & MAPPING: View Dynamic Product Map, Record Item Location, Contribute Location Data, Suggest Best Supermarket, Get Optimal Route (TSP), Navigate to Item in Store, Alert Near Item Location

ADMINISTRATION: Manage Users, Moderate Location Data, Map Analytics

Relationships:

- Mark Item Completed <<includes>> Record Item Location

- Record Item Location <<includes>> Contribute Location Data

- Contribute Location Data <<includes>> View Dynamic Product Map

- Get Optimal Route <<includes>> View Dynamic Product Map

- Suggest Best Supermarket <<includes>> View Dynamic Product Map

- Share List <<extends>> Invite Collaborator

- Edit Shared List <<extends>> Share List

- Navigate to Item <<extends>> Get Optimal Route

- Alert Near Item Location <<extends>> Navigate to Item in Store

Actor associations:

- Guest User → Manage Account

- Registered User → Manage Account, Manage List, Set Item Recurrent, View Shopping History, Share List, Invite Collaborator, Suggest Best Supermarket

- Active Shopper → Mark Item Completed, Record Item Location, View Dynamic Product Map, Get Optimal Route, Navigate to Item in Store, Alert Near Item Location

- List Collaborator → Edit Shared List, Manage List

- GPS/Location Service → Record Item Location, Navigate to Item in Store

- Map Aggregation Server → Contribute Location Data, View Dynamic Product Map, Get Optimal Route, Suggest Best Supermarket, Moderate Location Data

- Notification Service → Alert Near Item Location

- Administrator → Manage Users, Moderate Location Data, Map Analytics

it should also add for guest the use case to login logout and to be able to see the list but not edit it

-Gemini (for class diagram)

Prompt used:”ACTORS #ActorTypeDescription1Guest UserPrimaryUnregistered visitor; can register or log in2Registered UserPrimaryCore actor — manages lists, profile, preferences3Active ShopperPrimary (extends Registered User)Currently in-store with GPS/Wi-Fi RTT active4List

CollaboratorPrimaryInvited user co-editing a shared list in real-time5AdministratorPrimaryManages platform, users, location data quality6GPS / Location ServiceSecondary (System)Provides real-time coordinates via GPS or Wi-Fi RTT7Map Aggregation ServerSecondary (System)Collects & clusters item location data from all users8Notification ServiceSecondary (System)Sends push alerts and route updates USE CASES #Use CaseActor(s)CategoryRelations1Manage AccountGuest, Registered UserAuth2Manage ListRegistered UserList Mgmt(Create / Add-Edit-Remove / Delete)3Set Item RecurrentRegistered UserList Mgmt4View Shopping HistoryRegistered UserList Mgmt5Mark Item CompletedActive ShopperList Mgmt«includes» UC66Record Item LocationActive Shopper, GPS ServiceP2P«includes» UC77Contribute Location DataActive Shopper, Map ServerP2P8Share ListRegistered UserCollaboration«extends» UC99Invite CollaboratorRegistered UserCollaboration10Edit Shared List (RT)List CollaboratorCollaboration«extends» UC811View Dynamic Product MapActive ShopperP2P12Suggest Best SupermarketRegistered User, Map ServerP2P«includes» UC1113Get Optimal Route (TSP)Active Shopper, Map ServerP2P«includes» UC1114Navigate to Item in StoreActive Shopper, GPS ServiceP2P«extends» UC1315Alert Near Item LocationActive Shopper, Notification ServiceP2P«extends» UC1416Manage UsersAdministratorAdmin17Moderate Location DataAdministrator, Map ServerAdmin18Map AnalyticsAdministratorAdmin Ai aceste usecases si actors, fa mi o diagrama de clase astfel incat sa includa si sa excluda ce e necesar”

Advantages:

- Faster requirement structuring
- Helped identify relationships (aggregation/composition)
- Suggested <<include>> and <<extend>>

Disadvantages:

- Needed manual refinement
- The AIs misunderstood parts of the assignment